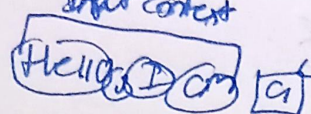
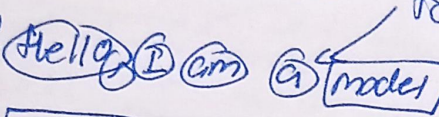
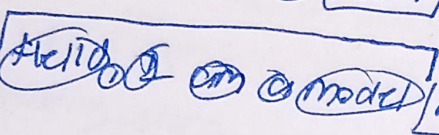


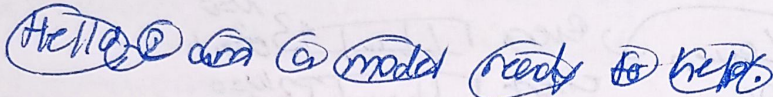
Generating text from output tensors!

↳ Review of how an LLM generates text

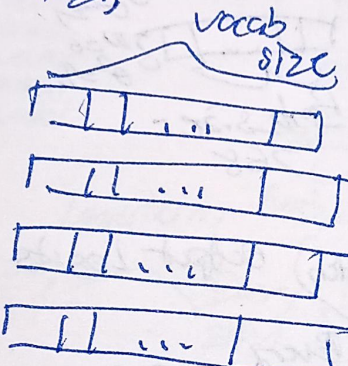
1st iter: 
Input context: Hello, I am
next generated token: a

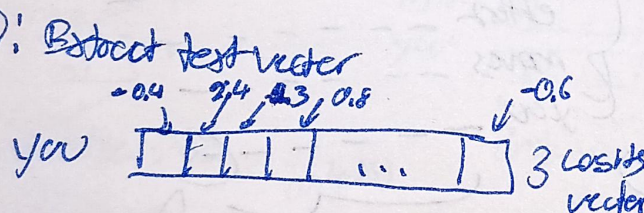
2nd iter: 
Token generated in the previous round is appended to the input for next iteration

3rd iter: 
Input context grows in each iteration

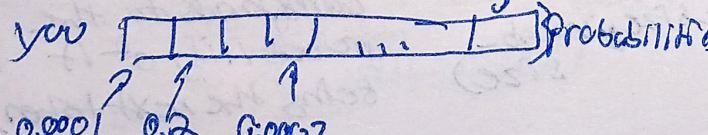
...
8th iter: 

* As per previous section,

Step (1): Every 
vocab size

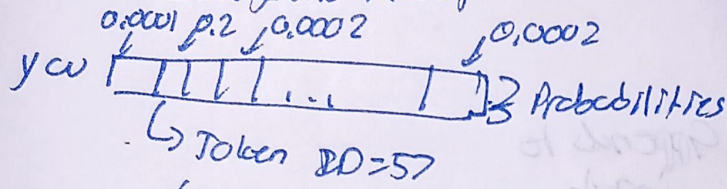
Step (2): Extract test vector 
-0.4, 2/4, 2/3, 0.8, ..., -0.6
3 losses vector

Step (3): convert losses into probabilities by applying softmax


0.0001, 0.2, 0.0002, ...
3 probabilities

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Step (4): Identify the index position (token ID) of largest value.

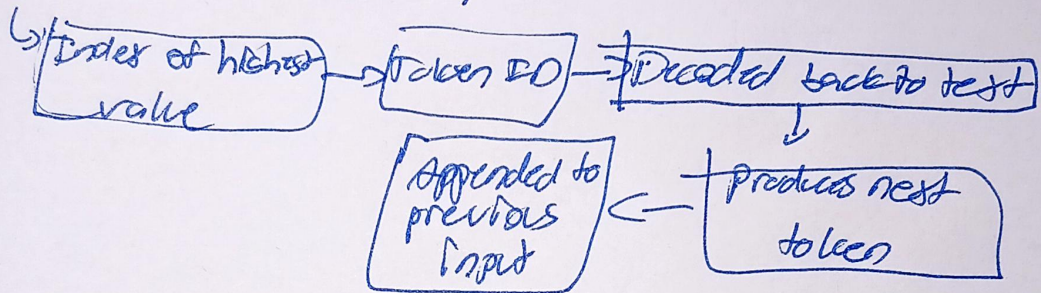


↳ Decode token ID to text:

57: "forward"

Step (5): Append token ID to the previous inputs for the next round

* Essentially, the vector corresponding to the next token is extracted and converted into a probability distribution via the softmax function.



* This step-by-step process enables the model to generate text sequentially, building coherent phrases from initial input context

↳ In practice, we repeat this process over many iterations until we reach a user specified # of generated tokens.

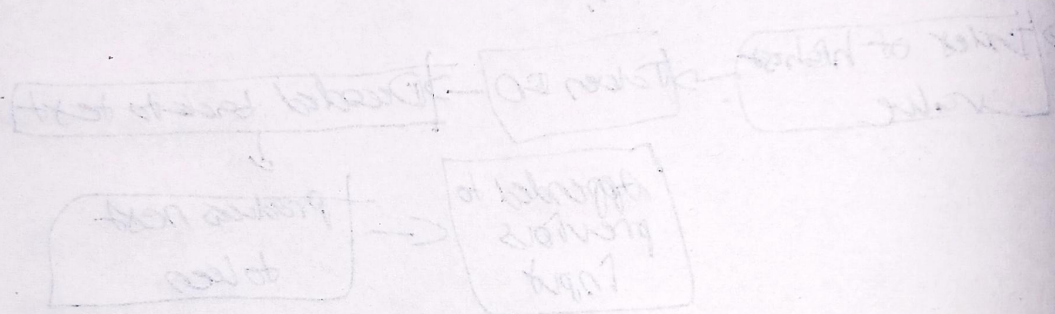
Iteration			
1	[15496, 11, 314, 716] Hello, I am	Predict	2D [257] a
2	[15496, 11, 314, 716, 257] Hello, I am a	append	
3	[15496, 11, 314, 716, 257, 2746] Hello, I am a model	model	[2746] ready
...			
6	[15496, ..., 3492, 284, 1632, 13] Hello, I am a model ready to help.		

The output tokens after 6 iterations (only new tokens = 6)

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PAGE

On Iteration 1, model is provided with tokens
to "Hello, I am"
↓
Predicts next token → Appends to
with ID 257
(61) input

This process is repeated until the model produces the
complete sentence: "Hello, I am a model
ready to help." after 6 iterations



Step 1 - Step 6: Repeat the model to generate
completing the sentence. The model generates
the next token. The model generates the next token.
The model generates the next token. The model generates the next token.
The model generates the next token. The model generates the next token.

Iteration	Input	Model Output
1	Hello, I am	[ID 257]
2	Hello, I am [ID 257]	[ID 258]
3	Hello, I am [ID 257] [ID 258]	[ID 259]
...	...	...
6	Hello, I am [ID 257] [ID 258] [ID 259] [ID 260] [ID 261] [ID 262]	[ID 263]